VINCENT WAMBUGU WAMBUI

Junior Python Developer | Backend and Full Stack Software Engineer

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PROFESSIONAL SUMMARY

Junior Python Developer and Computer Science graduate with more than one year of hands on software development, building and deploying production systems with Python (FastAPI, Django REST Framework and Flask), PostgreSQL, Redis, Docker and React front ends. Uses AI powered tools, mainly Claude, as part of a regular development workflow for design, debugging, test writing and documentation, and reviews every output critically before it reaches the codebase. Has also built AI features into products with the Claude, Gemini and Groq APIs, including retrieval augmented generation and validated model output. Writes clear, precise English in documentation, tickets and code, and learns quickly alongside experienced engineers. Currently building PulseOps (Python) and EventForge (Go), two reliability focused projects driven by automated testing. Available to start immediately.

TECHNICAL SKILLS

Programming: Python (FastAPI, Flask, Django REST Framework, SQLAlchemy, Alembic, Pydantic, pandas, httpx, Celery), SQL, JavaScript, TypeScript, Go, Node.js

Web and APIs: REST APIs, JSON, Swagger and OpenAPI, React, Next.js, Tailwind CSS, JWT, OAuth 2.0, role based access control, webhooks, third party API integration (M-Pesa Daraja, KCB Buni, Google OAuth)

Databases: PostgreSQL, MySQL, MongoDB, Redis, ClickHouse, pgvector, indexing and query optimisation, schema migrations

AI Powered Tools and AI Development: Claude (primary AI assistant), GitHub Copilot, ChatGPT, Gemini, Claude API, Gemini API, Groq API, prompt design, retrieval augmented generation, vector search, LLM output validation

Engineering Practices: Git and GitHub, develop to main workflow, Docker and Docker Compose, Render, Vercel, Django tests, property tests, API documentation, encrypted data and audit logging, debugging production failures

Languages: English (excellent written and spoken)

SELECTED PROJECTS

PulseOps: Evidence First Incident Investigator | In progress | github.com/VINN5

Python, telemetry analysis, deterministic scoring, citation checking, property tests

- Building a decision aid for OpenTelemetry instrumented services in Python: a deterministic pipeline detects metric shifts, aligns them with deploys and config changes, finds the most degraded upstream service from the dependency graph, and ranks likely causes.
- Built a naive baseline (blame the latest change) and a decoy deploy scenario where it fails and the pipeline holds, with a command line tool that produces reports from captured telemetry folders.
- Built a capped, ID tagged evidence bundle and a citation checker that rejects fabricated IDs, uncited claims, wrong evidence types, impossible time order and queries that are not allow listed; property tests confirm it never accepts a fake ID. 80 tests pass on main.

EventForge: Durable Job and Event Engine on PostgreSQL | In progress, milestone 3 of 6 | github.com/VINN5

Go, PostgreSQL, SKIP LOCKED, leases and fencing tokens, transactional outbox

- Building a job queue for small teams whose background work must survive crashes, with at least once delivery and idempotent handlers, and every guarantee tied to a named failure test.
- Implemented job claiming with SELECT FOR UPDATE SKIP LOCKED, lease based ownership with heartbeats, a reaper that requeues expired leases, and fencing tokens so a stale worker cannot overwrite a newer result.
- Implemented retries with exponential backoff and full jitter, retryable and permanent error classes and a dead letter store; now building idempotency keys, a transactional outbox and audited replay.

TapeLine: Multi Database Data Connector Platform | Live: the-tapeline.vercel.app

Python, Django REST Framework, Next.js, TypeScript, PostgreSQL, Celery, Redis, Docker, Render, Vercel

- Built connectors for PostgreSQL, MySQL, MongoDB and ClickHouse with configurable batch extraction, a query builder with column, operator and value filters plus sorting, and an inline editable data grid.
- Moved extraction to Celery and Redis background jobs so the API responds instantly while the frontend polls job status, and cut database queries per job from N to 1 using composite indexes, select_related and bulk_create.
- Secured the API with JWT and refresh token blacklisting, Fernet encrypted database credentials, role based access and authenticated downloads; exported edited data as JSON, CSV or styled Excel.

ReliefIQ: AI Knowledge and Governance Assistant for NGOs | github.com/VINN5/ReliefIQ
Python, FastAPI, PostgreSQL, pgvector, React, TypeScript, Gemini, Groq, Anthropic APIs

- Built a retrieval augmented Q&A assistant that answers only from an organisation's documents, using sentence transformer embeddings and pgvector search with High, Medium and Low confidence scoring.
- Enforced citations on the server so answers without verifiable citations are discarded, and escalated sensitive questions (safeguarding, legal, HR) to a named human.

Banalyst: AI Business Analyst for SMEs | In development
Python, FastAPI, SQLAlchemy, pandas, React, Tailwind CSS, Anthropic Claude API

- Separated a deterministic pandas rules engine from Claude narration so every insight rests on computed numbers; built CSV ingestion with fuzzy column mapping and partial success row handling.

MatSched (MatatusGo): Matatu Seat Booking and Fleet Platform | 2026 to Present
Python, FastAPI, React, TypeScript, PostgreSQL, Redis, Alembic, M-Pesa Daraja

- Designed a RESTful API with 20+ endpoints for four user roles, with JWT authentication, phone OTP, Google OAuth 2.0 and server side role based access.
- Implemented M-Pesa Daraja payment reconciliation with a commission ledger, plus encrypted personal data and audit logging for Data Protection Act compliance.

Summit Wheels: Car Rental Marketplace with Payments | Deployed
Python, FastAPI, React, TypeScript, MongoDB, Prisma, KCB Buni, M-Pesa STK Push, Render

- Built a two stage payment flow with KCB Buni and M-Pesa STK Push, including webhook callbacks, asynchronous processing and status polling, and diagnosed live production failures involving token handling, CORS and timeouts.

AgroWeather Nyeri: Weather Intelligence Dashboard for Farmers | Live demo, v1.0.0 release
Python, FastAPI, httpx, Pydantic v2, React 18, TypeScript, Vite, Tailwind CSS, Docker, Nginx

- Built a layered FastAPI backend (endpoints, services, schemas, config) with an async HTTP client that integrates a weather AI API for forecasts, canopy analysis and crop disease detection from uploaded photos.

Other projects | github.com/VINN5

- StockFlow: multi tenant inventory and point of sale system for Kenyan retail (Python, Flask, MongoDB Atlas) with role based access, client credit tracking, low stock alerts, profit analytics and Excel import and export.
- ORRBIT: multi domain AI assistant (Python, React) with graph based routing, three layer memory, web and arXiv retrieval and an input safety guard.

PROFESSIONAL EXPERIENCE

IT Support Intern | Astryx Dynamics, Nairobi | May 2025 to Aug 2026

- Primary on site technical contact: logged every incident in JIRA, resolved to SLA and escalated with full diagnostic detail, building a habit of precise written communication.
- Built a searchable knowledge base from support tickets and resolutions, and kept SOPs, configuration records and change diaries current for audit readiness.

Network Technician and IT Support | Ecospot Internet Ltd | Nov 2024 to Apr 2025

- Deployed LAN and WLAN setups for 50+ client sites and traced faults with Wireshark and link testing, reaching 95%+ client satisfaction.

ICT Assistant | NHIF Office, Kerugoya | May 2024 to Aug 2024

- Maintained LAN, workstations and the IT asset register under government security policy, trained staff and wrote network diagrams and incident logs.

ICT Support Intern | Ministry of Transport and Public Works, Nyeri | May 2023 to Aug 2023

- Supported connectivity for 80+ workstations and set up automated backups while digitising departmental records.

EDUCATION

BSc Computer Science | Kirinyaga University | Graduated 2025
Coursework: Software Development, Database Management, Distributed Systems, Cloud Computing, Network Security, Computer Networks.

REFEREES

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